BITS F464 Machine Learning — Assignment 1: Results Matrix

Anuran Calls (MFCCs) · stratified 80/20 split, seed 42 · 5,756 train / 1,439 test rows

Table 2: Results matrix (flat models predict Species; † see note)

Method	Criterion	Overall accuracy (Species) %		Family acc. † %
		Train	Test	Test
1. Single Decision Tree	Gini	100.00	94.93	96.25
	Entropy	100.00	95.41	96.80
2a. Bagging	Gini	99.97	96.66	97.50
	Entropy	99.98	96.87	97.50
2b. AdaBoost	Gini	97.52	95.34	96.73
	Entropy	96.35	94.16	95.00
3. Random Forest	Gini	100.00	97.78	98.40
	Entropy	100.00	97.64	98.19
4. Hierarchical Decision Tree	Gini	100.00	93.82	**Router: 95.07**
	Entropy	100.00	94.65	**Router: 95.62**
5. Gaussian Naive Bayes	—	91.47	90.34	91.94

† For flat models, the derived family accuracy (each predicted species mapped to its family via the species→family map read from the training data). For the Hierarchical rows, the router accuracy; their 'Overall accuracy' columns hold the final species accuracy.

Table 3: Supplementary — hierarchical failure analysis (test set, %)

Criterion	Router acc.	Oracle acc.	(A) Router right, expert wrong	(B) Router wrong
Gini	95.07	98.26	1.25	4.93
Entropy	95.62	98.33	0.97	4.38

Check: overall accuracy = router acc. − (A), and (B) = 100 − router acc., for both criteria.